EXPERIMENT 1:

```
import cv2 import
matplotlib.pyplot as plt

cb_img=cv2.imread("Downloads/download.jpg",cv2.IMREAD_GRAYSCA
LE) print(cb_img) plt.imshow(cb_img,cmap='gray')

plt.axis('off')
plt.show()
```

```
[[ 67  67  68 ...  59  59  59]
 [ 67  68  68 ...  59  59  59]
 [ 68  68  68 ...  59  59  59]
 ...
 [141 144 147 ...  98  98  98]
 [141 144 147 ...  94  94  94]
 [141 143 146 ...  93  93  93]]
```

```
print(cb_img) print("Shape of the
image:",cb_img.shape) print("Data type
of the image:",cb_img.dtype) print("Size
of the image:",cb_img.size)
plt.imshow(cb_img,cmap='gray')

plt.axis('off')
plt.show()
```

```
[[ 67  67  68 ...  59  59  59]
 [ 67  68  68 ...  59  59  59]
 [ 68  68  68 ...  59  59  59]
 ...
 [141 144 147 ...  98  98  98]
 [141 144 147 ...  94  94  94]
 [141 143 146 ...  93  93  93]]
Shape of the image: (148, 204)
Data type of the image: uint8
Size of the image: 30192
```

```
img=cv2.imread("Downloads/download.jpg")
print("Image shape (height,width,channels)
is:",img.shape) print("Image size is:",img.size)
print("Data type of image is:",img.dtype)
```

```
Image shape (height,width,channels) is: (148, 204, 3)
Image size is: 90576
Data type of image is: uint8
```

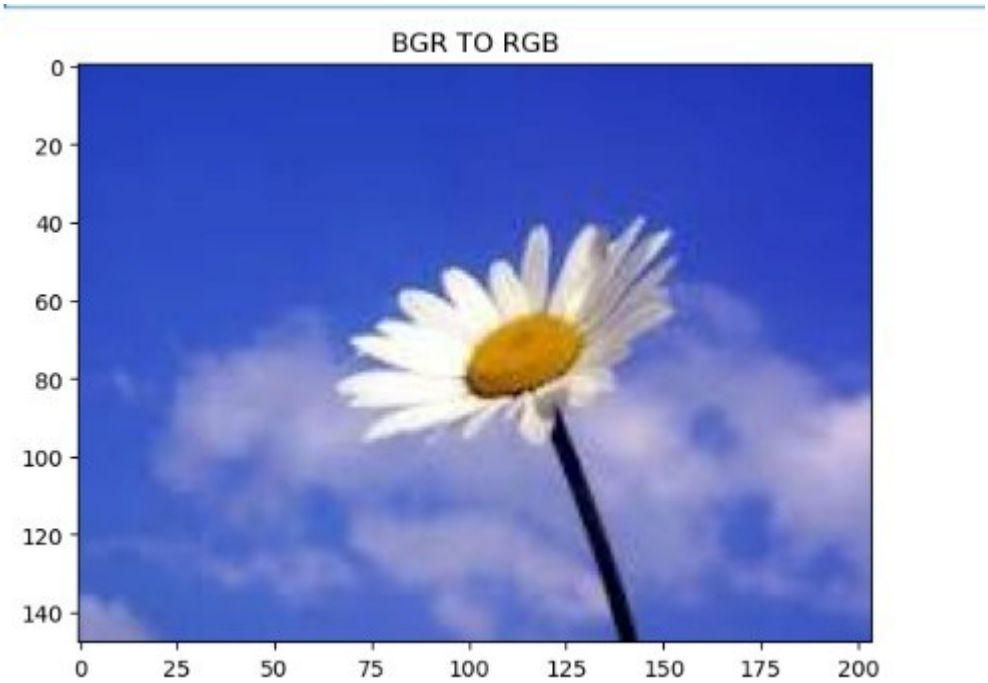
```
flower=cv2.imread("Downloads/download.jpg",cv2.IMREAD_COLOR)
plt.imshow(flower)

plt.axis('off')
plt.show()
```

```
reverse=flower[:,
:,::-1]
plt.imshow(rever
se) plt.axis('off')
plt.show()
```

```
b,g,r=cv2.split(flower)
plt.figure(figsize=(15,5))
plt.subplot(1,3,1)
plt.imshow(r,cmap="gra
y") plt.title("Red
channel") plt.axis('off')
plt.subplot(1,3,2)
plt.imshow(g,cmap='gra
y') plt.title("Green
```

Channel") plt.axis('off')
plt.subplot(1,3,3)
plt.imshow(b,cmap='gra
y') plt.title("Blue
Channel") plt.axis('off')
plt.show()

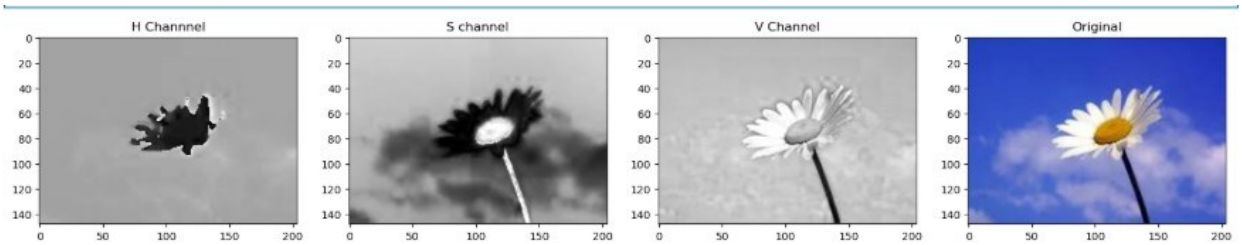
flower=cv2.imread("Downloads/download.jpg",cv2.IMREAD_COLOR)
flower1=cv2.cvtColor(flower,cv2.COLOR_BGR2RGB) plt.imshow(flower1) plt.title("BGR TO RGB") plt.show()



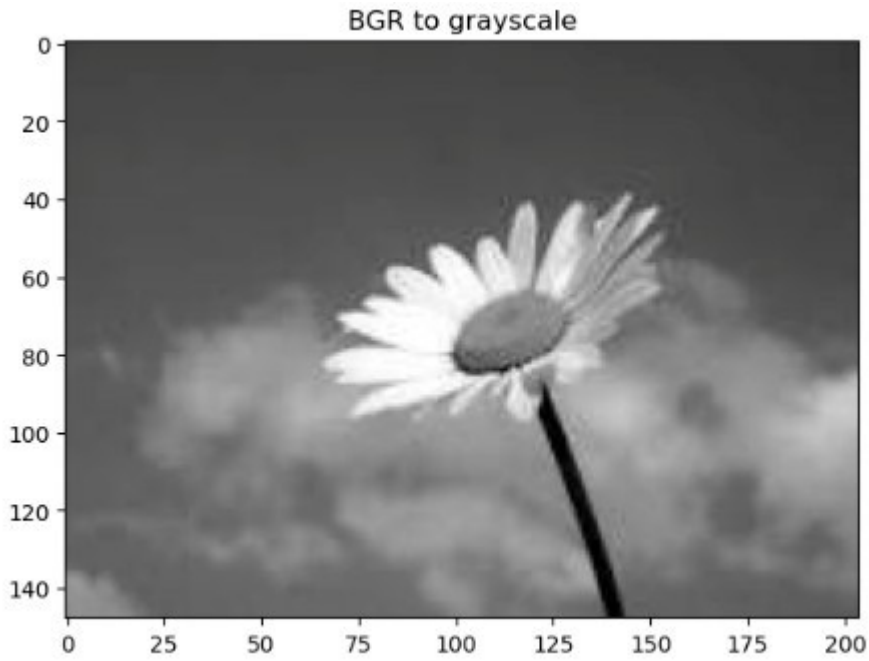
hsv=cv2.cvtColor(flower,cv2.COLOR_BGR2HSV) h,s,v=cv2.split(hsv)
plt.figure(figsize=[20,5]) plt.subplot(1,4,1) plt.imshow(h,cmap="gray")
plt.title("H Channnel")

plt.subplot(1,4,2) plt.imshow(s,cmap="gray")
plt.title("S channel") plt.subplot(1,4,3)
plt.imshow(v,cmap="gray") plt.title("V Channel")
plt.subplot(1,4,4)

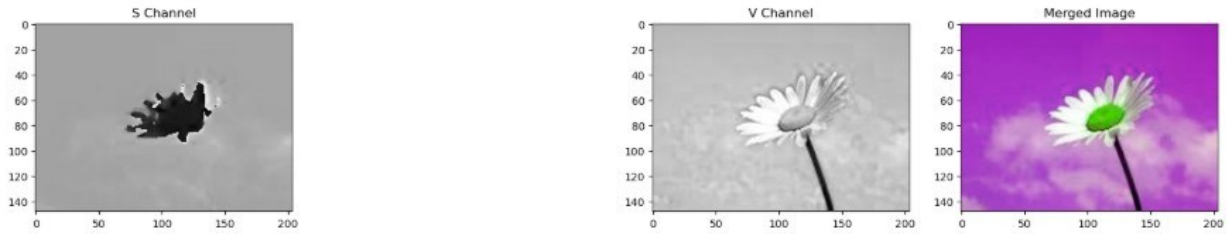
```
plt.imshow(cv2.cvtColor(flower,cv2.COLOR_BGR2RG
B)) plt.title("Original") plt.show()
```




```
flower=cv2.imread("Downloads/download.jpg",cv2.IMREAD_COLOR)
flower1=cv2.cvtColor(flower,cv2.COLOR_BGR2GRAY) plt.imshow(flower1,cmap="gray")
plt.title("BGR to grayscale") plt.show()
```




```
flower=cv2.imread("Downloads/download.jpg",cv2.IMREAD_COLOR)
hsv=cv2.cvtColor(flower,cv2.COLOR_BGR2HSV)

h,s,v=cv2.split(hsv) h_new=h+30
flower_merged=cv2.merge((h_new,s,v))
flower_rgb=cv2.cvtColor(flower_merged,cv2.COLOR_HSV2R
GB) h,s,v=cv2.split(flower_merged) plt.figure(figsize=[20,5])
plt.subplot(1,4,1) plt.imshow(h,cmap="gray") plt.title("S
Channel") plt.subplot(1,4,3) plt.imshow(v,cmap='gray')
plt.title("V Channel") plt.subplot(1,4,4)
plt.imshow(flower_rgb) plt.title("Merged Image") plt.show()
```




```
flower=cv2.imread("Downloads/download.jpg",cv2.COLOR_BGR2RGB)
cv2.imwrite("saveflower.jpg",flower1)
```